

JAVA HIERARCHICAL INHERITANCE PROGRAM

Object-Oriented Programming Practical Record

1. Aim

To write a Java program using hierarchical inheritance to calculate the area of different shapes such as a rectangle and a circle.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named Main.java.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create objects for Rectangle and Circle.
6. Rectangle inherits the message() method from Shape.
7. Circle inherits the message() method from Shape.
8. Call the message() and area() methods for both objects.
9. Display the calculated areas.

3. Algorithm

1. Start the program.
2. Create the parent class Shape.
3. Create message() method in Shape to display a message.
4. Create the Rectangle class extending Shape.
5. Declare length and width in Rectangle.
6. Create area() method in Rectangle.
7. Create the Circle class extending Shape.
8. Declare radius in Circle.
9. Create area() method in Circle.
10. Create objects of Rectangle and Circle in main().
11. Call message() and area() for both objects.
12. Display the calculated areas.
13. Stop the program.

4. Explanation of the Program

Shape class: It is the parent class containing the message() method, which displays "Calculating Area".

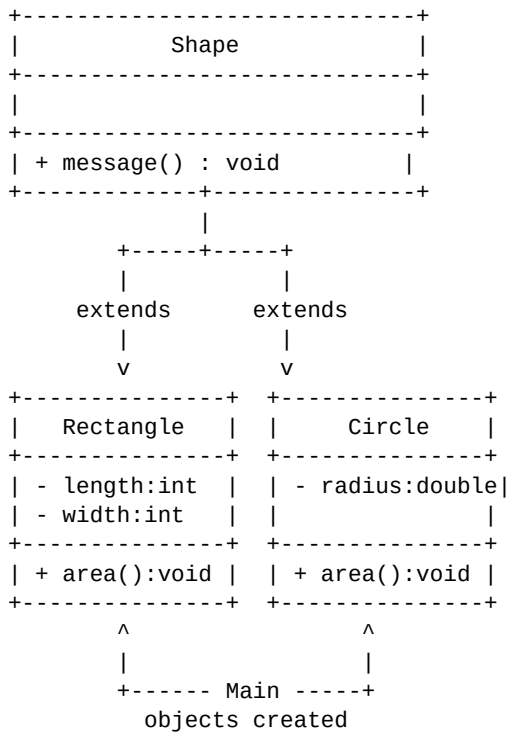
Rectangle class: It is a child class that extends Shape. It inherits message() and provides the area() method for calculating rectangle area.

Circle class: It is another child class that extends Shape. It inherits message() and provides the area() method for calculating circle area.

Hierarchical Inheritance: The program demonstrates hierarchical inheritance because Rectangle and Circle both inherit from the same parent class, Shape.

Main class: The main() method creates Rectangle and Circle objects and calls their inherited and own methods.

5. UML Class Diagram



6. Java Source Code

```
// Hierarchical Inheritance
// Real application: Area Calculator

class Shape {
    void message() {
        System.out.println("Calculating Area");
    }
}

class Rectangle extends Shape {

    int length = 10;
    int width = 5;

    void area() {
        System.out.println("Rectangle Area = " + (length * width));
    }
}

class Circle extends Shape {

    double radius = 7;

    void area() {
        System.out.println("Circle Area = " + (3.14 * radius * radius));
    }
}

public class Main {
    public static void main(String[] args) {

        Rectangle r = new Rectangle();
        r.message();
        r.area();

        Circle c = new Circle();
        c.message();
    }
}
```

```
        c.area();  
    }  
}
```

7. Output of the Program

```
Calculating Area  
Rectangle Area = 50  
Calculating Area  
Circle Area = 153.86
```

8. Result

Thus, the Java program successfully demonstrates hierarchical inheritance using a real-world area calculator application. The Rectangle and Circle classes inherit the message() method from Shape and calculate their respective areas.